

HW04 Automation Testing Report

Document control

Field	Value
Student	Lê Tuấn Lộc
Student ID	23127404
Class	PENDING — enter official class code
Repository	https://github.com/KidCute1412/eshop-sut
Branch inspected	23127404-LeTuanLoc
Report status	Nine-run execution evidence reconciled; manual submission artifacts pending
Final submission timestamp	PENDING — ISO 8601
Execution revision	656991a598bafe43cbcd13b54bf1ac3f429c30f2 ; recorded in every selected report metadata file

Executive summary

HW04 requires data-driven automation for the same three web features selected in HW02: FR-06 Product detail view, FR-10 Order state machine, and FR-12 Access control. Each feature requires at least 12 cases and execution on three browsers, producing at least nine attributable HTML reports overall.

The deliverables suite defines 51 external-data cases: 16 for FR-06, 16 for FR-10, and 19 for FR-12. All cases were scheduled on Chromium, Firefox, and WebKit in nine selected feature/browser runs. The nine complete HTML/JSON/metadata report directories passed structural identity, timestamp, case-count, completion, and totals validation. Across 153 browser attempts, 102 reached assertions: 64 passed and 38 failed across 19 unique logical cases. The other 51 attempts failed during Firefox `browserContext.newPage()` fixture setup before the test body could observe the SUT.

FR-06 produced 20 passes and 28 recorded failures across 48 executions; 16 failures were environmental and 12 were assertions. FR-10 produced 24 passes and 24 recorded failures across 48 executions; 16 were environmental and 8 were assertions. FR-12 produced 20 passes and 37 recorded failures across 57 executions; 19 were environmental and 18 were assertions.

Firefox was run headful after headless Firefox failed before `newPage()` on the assignment Windows environment. The headful final runs remained intermittently affected by `newPage()` timeouts; those failures are not counted as SUT defect candidates. The Markdown/PDF report set is complete. No public HW04 Issue URL or demo video is currently verified, and the four-day Git-history span remains noncompliant.

The full HW04 Git history contains nine qualifying test-script commits across 27 July and 9 August 2026, so the commit-count minimum is met. The separate four-calendar-day requirement remains noncompliant. This report does not infer, manufacture, or backdate missing history.

1. Scope and test oracle

Pool	Requirement	Observable obligation	Source continuity
A	FR-06 — Product detail view	Show image, name, formatted price, description, and category; accept only positive integer quantity with minimum 1; provide visible add-to-cart feedback	HW02 FR-06 design and root <code>README.md</code>
B	FR-10 — Order state machine	Enforce permitted transitions across <code>pending</code> , <code>confirmed</code> , <code>shipping</code> , <code>delivered</code> , and <code>canceled</code> ; preserve final states and role restrictions	HW02 FR-10 design and root <code>README.md</code>
C	FR-12 — Access control	Restrict Admin UI, <code>/api/admin/*</code> , and data-changing product/category/coupon APIs to a valid JWT with <code>role = 'admin'</code>	HW02 FR-12 design and root <code>README.md</code>

The final 51-row matrix is in `../supporting-materials/test_case_matrix.md`, with IDs `FR06-01 – FR06-16`, `FR10-01 – FR10-16`, and `FR12-01 – FR12-19`. It maps every actual external-data row to its spec and to the Chromium, Firefox, and WebKit outcome/report path.

2. Test design and traceability

2.1 Coverage targets

Feature	Planned IDs	Positive	Negative/boundary/edge focus	Automated now	Final evidence
FR-06	Source <code>FR06-01 – FR06-16</code>	Existing products and valid quantity boundaries	Content/format, quantity constraints, feedback, unknown ID	16 cases × 3 browsers	48 attempted; 32 reached assertions: 20 pass, 12 fail; 16 fixture failures
FR-10	Source <code>FR10-01 – FR10-16</code>	Permitted admin/user transitions	Skipped transitions, role limits, and final-state exits	16 cases × 3 browsers	48 attempted; 32 reached assertions: 24 pass, 8 fail; 16 fixture failures
FR-12	Source <code>FR12-01 – FR12-19</code>	Authorized Admin UI/API operations	Guest/user/malformed authorization and protected mutations	19 cases × 3 browsers	57 attempted; 38 reached assertions: 20 pass, 18 fail; 19 fixture failures

2.2 Required row-level traceability

Every final case must map through the following chain:

Requirement → case ID → external data ID → test title/spec path → browser run ID → HTML report/evidence → defect/Issue when failed

The final matrix must record concrete preconditions, test data, steps, expected result, priority, automation decision, script path, and evidence reference. A case marked `Fail` must reference a defect or a documented non-SUT failure classification.

3. Automation implementation

3.1 Current inventory

Feature	Spec file	External data	Data-driven proof	Assertion patterns	Setup/cleanup	State
FR-06	../automation/tests/fr06-product-detail.spec.ts	../automation/test-data/fr06-product-detail.json	loadCases(..., 12) loads 16 rows	Text/content, visibility, count, attribute, value/validity	View cases plus cart-state-sensitive actions	Attempted in 3 selected reports
FR-10	../automation/tests/fr10-order-state-machine.spec.ts	../automation/test-data/fr10-order-state-machine.json	loadCases(..., 12) loads 16 rows	HTTP status/body, persisted state, visibility and button count	API driver arranges a fresh order state per row	Attempted in 3 selected reports
FR-12	../automation/tests/fr12-access-control.spec.ts	../automation/test-data/fr12-access-control.json	loadCases(..., 12) loads 19 rows	HTTP status/content type/body plus Admin/Login UI visibility	Disposable mutations with cleanup helper	Attempted in 3 selected reports

3.2 Current Playwright configuration

The deliverables suite pins Playwright 1.55.0 and defines Chromium, Firefox, and WebKit projects. It uses Playwright web-first assertions, workers: 1, CI-only retry, screenshots/traces/videos on failure, and a custom metadata reporter. Its report key includes feature, browser, and an ISO-derived timestamp; metadata includes Run by: 23127404, feature, browser, and timestamp.

The legacy bundle outside deliverables still produces one combined playwright-report directory. The new run:matrix workflow is intended to preserve distinct feature/browser report directories under the automation workspace. Each generated directory must still be opened and verified before it enters the final manifest.

3.3 Final implementation inventory template

Feature	Case count	Spec file(s)	Data file(s)	Locator strategy	Assertions	Isolation/reset	Reviewer approval
FR-06	16 source rows	../automation/tests/fr06-product-detail.spec.ts	../automation/test-data/fr06-product-detail.json	Roles plus scoped CSS where no stable semantic locator exists	Text, visibility, count, attribute, value/validity	Executed serially on 3 browsers	Human evidence triage recorded; final student sign-off pending

Feature	Case count	Spec file(s)	Data file(s)	Locator strategy	Assertions	Isolation/reset	Reviewer approval
FR-10	16 source rows	../automation/tests/fr10-order-state-machine.spec.ts	../automation/test-data/fr10-order-state-machine.json	API arrangement plus exact order-cell/row locators	Status/body, persisted state, visibility/count	Fresh arranged order per row	Substring locator corrected and all FR-10 runs repeated
FR-12	19 source rows	../automation/tests/fr12-access-control.spec.ts	../automation/test-data/fr12-access-control.json	API assertions plus Admin/Login headings	Status, content type/body, visibility	Cleanup helper for created probes	Executed; access-control failures retained

4. Execution and multi-browser evidence

4.1 Environment record

Field	Final value
Operating system	Windows 10 Home Single Language; Windows NT 10.0.26200.0
Node.js / npm	Node.js 22.17.0 ; npm 10.9.2
Playwright	1.55.0
Browser modes/versions	Chromium 140.0.7339.16 headless; Firefox 141.0 headful; WebKit 26.0 headless
Frontend URL	Current default http://localhost:5173 ; confirm final URL
Backend URL	Current default http://localhost:3000 ; confirm final URL
Execution revision/data seed	656991a598baf43cbcd13b54bf1ac3f429c30f2 ; reports generated from clean seed database and metadata records the revision/command
Execution operator	Lê Tuấn Lộc — 23127404

4.2 Historical repository evidence — excluded from final totals

Evidence	Repository-recorded value	Qualification
Logical case	FR06-DT-01	One view-only FR-06 case
Browsers represented	Chromium, Firefox, WebKit	Stored in one combined report
Recorded timestamp	2026-07-27T08:11:08.301Z	Taken from the existing student report/report metadata
Recorded result	0 passed, 3 failed	Report says each browser failed at the category assertion
Preserved report	../schools/23127404-automation/playwright-report/index.html plus assets	Historical only; not copied into the final nine-run structure
Defect status	Candidate only for this HW04 package	No verified HW04 public Issue URL recorded here

The preceding row is provenance-aware reporting of an existing artifact. It does not establish that the environment remains unchanged or that the current SUT still exhibits the behavior.

4.3 Final execution ledger

Run ID	Feature/browser	ISO timestamp	Attempted	Passed	Recorded failed	Environment failures within failed	Selected report directory
RUN-A-CHR	FR-06 / Chromium	2026-08-09T17:11:54.712Z	16	10	6	0	automation/reports/fr06-chromium-2026-08-09T17-11-54-712Z/
RUN-A-FF	FR-06 / Firefox	2026-08-09T17:13:19.066Z	16	0	16	16	automation/reports/fr06-firefox-2026-08-09T17-13-19-066Z/
RUN-A-WK	FR-06 / WebKit	2026-08-09T17:16:10.035Z	16	10	6	0	automation/reports/fr06-webkit-2026-08-09T17-16-10-035Z/
RUN-B-CHR	FR-10 / Chromium	2026-08-09T17:17:32.516Z	16	12	4	0	automation/reports/fr10-chromium-2026-08-09T17-17-32-516Z/
RUN-B-FF	FR-10 / Firefox	2026-08-09T17:18:43.660Z	16	0	16	16	automation/reports/fr10-firefox-2026-08-09T17-18-43-660Z/
RUN-B-WK	FR-10 / WebKit	2026-08-09T17:21:22.978Z	16	12	4	0	automation/reports/fr10-webkit-2026-08-09T17-21-22-978Z/
RUN-C-CHR	FR-12 / Chromium	2026-08-09T17:22:38.418Z	19	10	9	0	automation/reports/fr12-chromium-2026-08-09T17-22-38-418Z/
RUN-C-FF	FR-12 / Firefox	2026-08-09T17:28:39.697Z	19	0	19	19	automation/reports/fr12-firefox-2026-08-09T17-28-39-697Z/
RUN-C-WK	FR-12 / WebKit	2026-08-09T17:27:20.694Z	19	10	9	0	automation/reports/fr12-webkit-2026-08-09T17-27-20-694Z/
Total	3 features / 3 browsers	—	153 attempts	64	89	51	Nine structurally validated directories

The 89 runner failures equal 38 assertion failures plus 51 Firefox environment failures. All nine selected HTML/JSON/metadata report sets passed structural identity, timestamp, case-count, completion, report/data-directory presence, and totals validation. Each selected report metadata file records the exact `node scripts/run-matrix.mjs` command and revision `656991a598baf43cbcd13b54bf1ac3f429c30f2`.

The selected reports were generated from revision `656991a598baf43cbcd13b54bf1ac3f429c30f2`; the revision is embedded in every report metadata file and HTML attribution banner.

5. Human review and AI gap analysis

Human review retained strict requirement assertions, corrected one demonstrated locator defect, reran FR-10, and separated Firefox infrastructure symptoms from SUT behavior.

Feature/case	AI proposal or risk	Why insufficient	Human correction visible/documented	Evidence state
FR-06 / FR06-07	A happy-path check could omit category	FR-06 explicitly requires category; a reduced oracle could create a false pass	Exact category assertion was retained	Failed by assertion in all three final browser runs
FR-06 / locators	CSS selectors could couple tests to styling	Classes are not stable user-facing contracts	Roles, accessible names, and exact text are preferred; scoped CSS remains only for price styling where semantic hooks are absent	Visible in final spec and reports
FR-06 / data	Inline literals would violate data-driven requirement	Inline test arrays/objects are explicitly disallowed	Input and expected values were moved to external JSON	Visible in current spec/data
FR-06 / waits	Fixed sleeps could hide timing defects and cause flakiness	Duration is environment-dependent	Playwright assertions and auto-waiting are used	Visible in current spec
FR-10 / FR10-02	Initial row filtering used substring matching for an order ID	Order #2 could match rows such as #20, producing a false WebKit failure	Filter now requires an exact table cell for #<orderId>; all three FR-10 runs were repeated	FR10-02 no longer fails in the selected final Chromium, Firefox, or WebKit reports
Firefox environment	A browser failure could be misreported as a product defect	Headless Firefox failed before <code>newPage()</code> ; headful runs still had page-setup timeouts	Firefox configured headful; 51 <code>newPage()</code> timeouts classified as environment failures	Final Firefox JSON reports: 16 FR-06, 16 FR-10, 19 FR-12 environment failures
FR-12 / authorization assertions	Passing over status mismatches would hide security behavior	FR-12 requires valid admin JWT and role for protected endpoints	Expected 401/403 assertions were retained; actual 200/400 responses remain failures	18 assertion failures across nine logical cases in the selected reports

The mandatory critique is in `ai_critique.md`. The interaction record is in `ai_audit_report.md`.

6. Defects and issue tracking

The final runs produced 38 assertion failures across 19 unique logical cases: six FR-06, four FR-10, and nine FR-12 cases. These are documented as runtime-observed candidates in `./bugs/bug_report.md`; the 51 Firefox `newPage()` failures are documented separately as environment failures. No candidate is represented as a filed GitHub defect because no verified public HW04 Issue URL or attached public screenshot has been supplied.

For every final failure:

1. complete case-level triage and rule out any remaining locator, assertion, data, authentication, state-reset, browser-installation, or infrastructure fault;
2. retest once when safe;
3. preserve the report, trace, and authentic screenshot;
4. create one deduplicated GitHub Issue for the root behavior; and
5. reconcile case IDs, run IDs, evidence references, and Issue URL in both bug files.

Prior HW02/HW03 issues may support regression context but are not evidence that a new HW04 Issue was filed or verified.

7. AI use

Required declaration: **I use AI tools for the following tasks**, preparing automation and submission documentation, reviewing traceability and evidence integrity, and drafting structured templates. The surviving audit history is incomplete: one historical exact prompt was not retained, and that omission is disclosed rather than reconstructed. See `ai_audit_report.md`.

8. Git history

The exported log in `../git/23127404_HW04_git_commit_log.txt` contains nine qualifying `.spec.ts` commits: legacy `f905546` on 27 July 2026 plus eight deliverables-suite commits from `f732b2c` through `aa316e0` on 9 August 2026. The commit minimum is met across two calendar days. The separate four-day requirement remains **noncompliant**; documentation commits do not change the qualifying span, and the shortfall must not be hidden through backdating or altered timestamps.

9. Demonstration videos and Agent Skill

Item	Requirement	Current status	Evidence
Main demo	Unlisted YouTube, at least 5 minutes, Vietnamese narration, end-to-end script, multi-browser run/report, one human fix, authorship evidence	Not recorded	<code>../video/demo_video.md</code>
Agent Skill	Reusable data-driven multi-browser workflow plus separate end-to-end video	Source present; end-to-end validation/demo not verified	<code>../agent-skill/playwright-data-driven-multibrowser/</code> ; <code>../video/agent_skill_demo.md</code>

The Agent Skills assessment category is currently claimed as 0/10. Complete source exists, but this must change only after a valid end-to-end demonstration is recorded and verified.

10. Limitations and remaining work

- The 51 cases have qualifying selected reports, but the exact execution commit and verbatim per-run shell commands were not embedded/retained. Environment and browser versions are recorded.
- The 19 assertion-failing cases require final root-cause deduplication, public GitHub Issues, and authentic Issue screenshots before they can be claimed as submitted defects.
- Firefox produced 51 `newPage()` environment failures even in headful mode; those cases were not observed by the SUT assertions.
- Historical and current AI logs still require complete prompt/output coverage for every generation/review interaction.
- Environment and browser versions are recorded; the exact execution revision remains unavailable because it was not embedded in the selected reports.
- No HW04 public Issue URL or final defect screenshot is verified.

- Both video links are absent; the main demo is mandatory, and Agent Skill points cannot be claimed from source alone.
- Historical AI logging lacks at least one exact prompt and timestamp.
- Nine qualifying commits are present across two days; the four-calendar-day history requirement remains unsatisfied.
- Markdown-to-PDF exports are complete and text-validated; the final archive must be regenerated after student-controlled URLs and identity fields are supplied.

11. Final reconciliation table

This table is reconciled from the nine selected final JSON reports and their environment triage.

Measure	FR-06	FR-10	FR-12	Total
Designed logical cases	16	16	19	51
Automated logical cases	16	16	19	51
Browser attempts	48	48	57	153
Attempts reaching assertions	32	32	38	102
Passed assertions	20	24	20	64
Recorded runner failures	28	24	37	89
Environment failures within recorded failures	16	16	19	51
Assertion failures	12	8	18	38
Unique assertion-failing logical cases	6	4	9	19
Verified public HW04 Issues	0	0	0	0

The arithmetic reconciles: $64 + 38 + 51 = 153$, $64 + 38 = 102$, and $38 + 51 = 89$. Issue counts must be updated only from verified public URLs.